E-COMMERCE DATA ANALYTICS

Final Project Report

Tools used:

- Excel
- PostgreSQL / SQL
- Power BI

Submitted by,
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1. Executive Summary

This project applies a practical data analytics workflow to an e-commerce transactions dataset. The work follows the three levels described in the Primeor Solutions internship brief: data understanding and cleaning in Excel, analytical querying with SQL, and an interactive three-page Power BI dashboard. The goal was to turn transaction-level records into a clear view of sales, profitability, customer activity, product performance and regional trends.

The cleaned working dataset contains **51,271 records** and **21 fields**, covering orders dated from **01 Jan 2011** to **12 Dec 2014**. The analysis reports **$12,635,222.00** in total sales and **$1,467,497.61** in total profit, with an average discount of **14.29%**. It contains **25,029** distinct orders, **3,788** distinct products and **796** distinct customers.

The dashboard is organized into an Executive Business Overview, Sales Analysis, and Product & Customer Insights. Slicers, KPI cards, interactive visuals, page navigation and a consistent blue/orange theme support exploration of the results.

Project snapshot

Measure	Result	Measure	Result
Total Sales	$12,635,222.00	Total Profit	$1,467,497.61
Distinct Orders	25,029	Average Discount	14.29%
Distinct Products	3,788	Unique Customers	796
Total Quantity	178,242	Average Order Value	$504.82

The results in this report are calculated from the cleaned Excel workbook available for this project. Values may display in rounded form on dashboard cards; the underlying measures retain the detailed totals.

2. Project Brief and Scope

The internship brief defines three project levels and a final submission package. This report documents the work against those instructions rather than introducing unrelated analysis.

Level 1 — Data Understanding & Cleaning (Excel)

Task 1: Clean the raw dataset by removing duplicate records, handling missing or null values, standardizing inconsistent text, correcting date formats, checking numerical values, and removing unnecessary spaces or formatting inconsistencies.

Task 2: Calculate total sales, total profit and average discount; identify the top five products by sales and the five most loss-making products; and analyze region-wise sales.

Task 3: Prepare pivot-style summaries and charts for sales by region, profit by category, segment-wise sales and monthly sales trend.

Level 2 — Data Analysis (SQL)

Task 1: Import the cleaned dataset into SQL. Task 2: write queries for top profitable products, top customers by sales, region-wise sales, category-wise average profit, highest-discount category, negative-profit orders, monthly sales trend, market-wise revenue, top sub-categories and shipping-mode usage. Task 3: summarize the business insights from those query results.

Level 3 — Power BI Dashboard

Task 1: Build three dashboard pages: Executive Overview, Sales Analysis, and Product & Customer Insights. Task 2: include slicers/filters, interactive visuals, KPI cards, proper formatting and a consistent colour theme.

3. Level 1 — Excel Data Understanding & Cleaning

The dataset was prepared as a structured analysis table before summaries and dashboard measures were created. The cleaned workbook contains 51,271 transaction rows and 21 columns, including order and shipping dates, customer, segment, geography, product, sales, quantity, discount, profit, shipping cost and order priority.

Cleaning and preparation performed

The preparation workflow addressed duplicate records, missing or null values, inconsistent text entries, date parsing, numerical fields and formatting consistency. Order and ship dates were converted to date values so that monthly and yearly analysis could be grouped correctly. Product, customer, region, market, segment and category fields were retained as analysis dimensions.

Excel analysis outputs

Analysis item	Result / finding
Total sales	$12,635,222.00
Total profit	$1,467,497.61
Average discount	14.29%
Top 5 products by sales	Apple Smart Phone, Full Size; Cisco Smart Phone, Full Size; Motorola Smart Phone, Full Size; Nokia Smart Phone, Full Size; Canon imageCLASS 2200 Advanced Copier
Top 5 loss-making products	Cubify CubeX 3D Printer Double Head Print; Lexmark MX611dhe Monochrome Laser Printer; Motorola Smart Phone, Cordless; Cubify CubeX 3D Printer Triple Head Print; Bevis Round Table, Adjustable Height
Region-wise sales leader	Central — $2,821,740
Highest-profit category	Technology — $663,200
Largest segment by sales	Consumer — $6,503,476

Pivot-style summaries and charts were prepared for region-wise sales, profit by category, segment-wise sales and the monthly sales trend. The workbook also includes separate sheets for the cleaned dataset, Task 2 analysis, top products, loss-making products and the summary tables used for charting.

4. Level 2 — SQL Analysis

The SQL stage is intended to make the cleaned records queryable and to answer business questions using aggregation, grouping, sorting and filtering. The project workflow used PostgreSQL through pgAdmin, with the cleaned e-commerce data organized in the `ecommerce_dataset` table within the `Ecommerce_db` database.

Query coverage and corresponding business questions

SQL analysis	Finding from the cleaned data
Top 10 profitable products	Canon imageCLASS 2200 Advanced Copier is the most profitable product, contributing $25,199.93 in profit.
Top 10 customers by sales	Tom Ashbrook leads customer sales with $40,489.
Region-wise total sales	Central is the highest-sales region ($2,821,740).
Category-wise average profit	Technology has the highest average profit per transaction ($65.46).
Highest discount category	Furniture has the highest average discount (16.80%).
Negative-profit orders	7,538 distinct orders contain at least one negative-profit row (12,538 negative-profit records).
Monthly sales trend	The strongest month in the period is 2014-12, with sales of $207,300.
Market-wise revenue	APAC is the highest-sales market ($3,581,785).
Top-performing sub-categories	Phones leads sub-category sales with $1,706,642.
Shipping-mode usage	Standard Class is the most frequently used mode (30,764 records).

The findings above are calculated from the cleaned dataset and summarize the outputs targeted by the SQL task list. They are suitable as validation points when reviewing the corresponding SQL query results.

Interpretation

The Central region and APAC market are the largest sales contributors in their respective geographic groupings. Technology is the largest contributor to total category profit, while Furniture has the highest average discount.

Standard Class is used most frequently, so shipping operations are concentrated in that service level. Negative-profit transactions are material enough to warrant a closer review by product, discount and shipping cost.

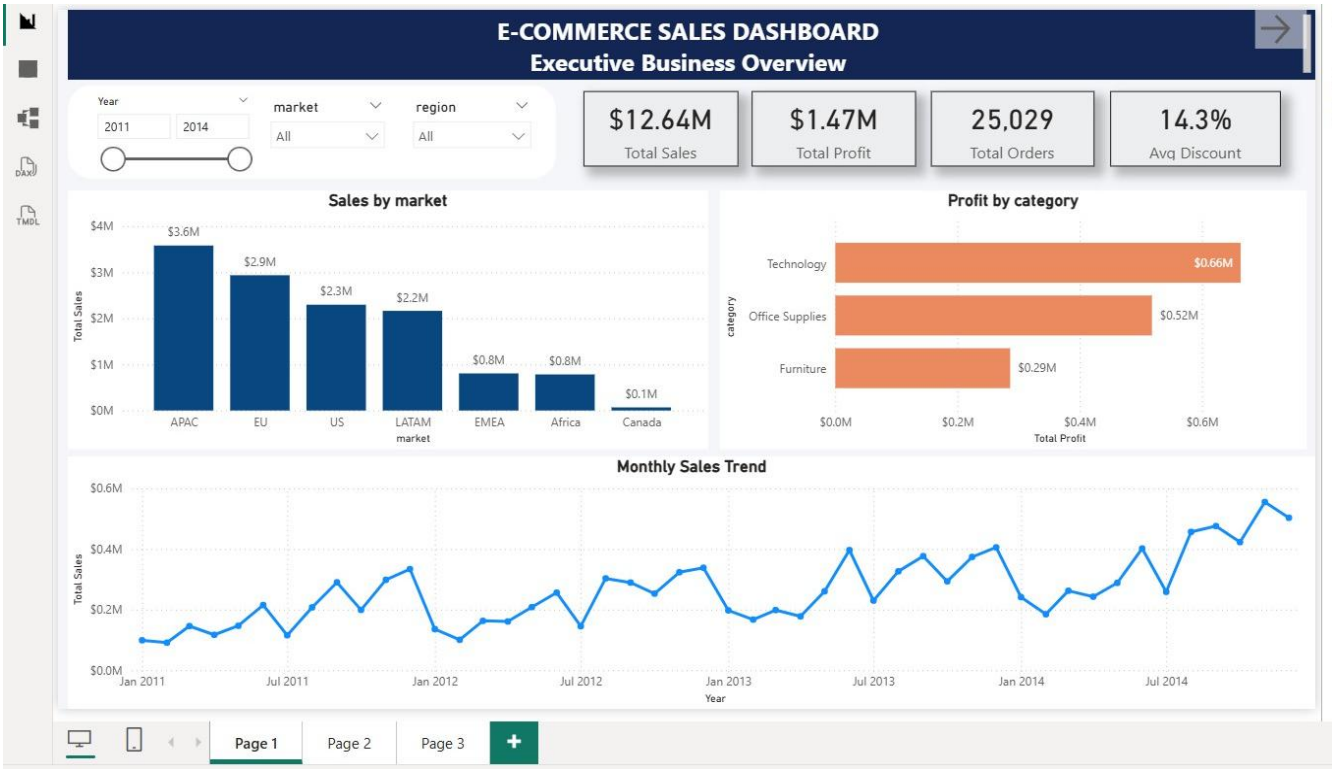
5. Level 3 — Power BI Dashboard

The Power BI report contains three pages built around the questions in the internship brief. Each page uses KPI cards, charts and slicers to allow users to move from a high-level summary into more detailed sales, product and customer analysis.

Page 1 — Executive Business Overview

The overview page presents Total Sales, Total Profit, Total Orders and Average Discount. Its visuals show Sales by Market, Profit by Category and Monthly Sales Trend. The page includes Year, Market and Region slicers, plus navigation to the other report pages.

Figure 1. Executive Business Overview dashboard page.



6. Power BI Dashboard — Sales and Customer Insights

Page 2 — Sales Analysis

The Sales Analysis page includes Total Sales, Total Orders, Total Quantity Sold and Average Order Value KPI cards. The four visuals are Monthly Sales Trend, Sales by Region, Sales by Market and Category Performance. Year, Market and Segment slicers support filtering.

Page 3 — Product & Customer Insights

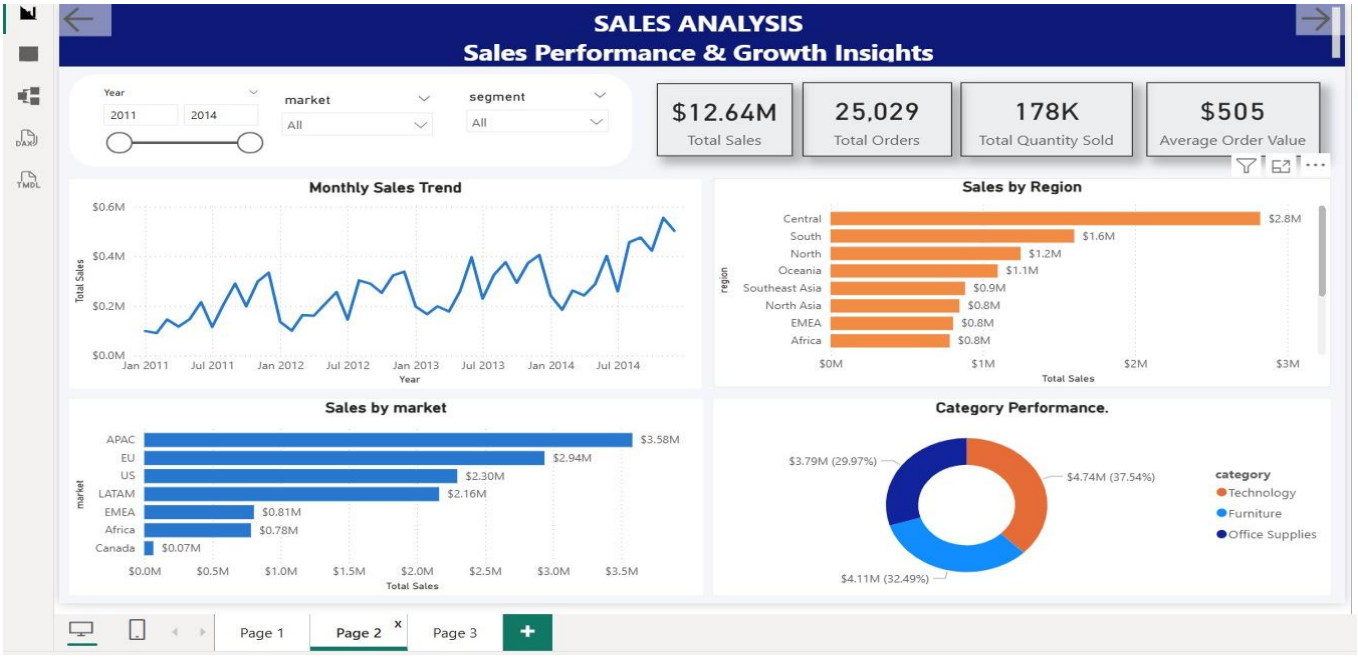


Figure 2. Sales Analysis dashboard page.

The third page contains Total Products, Unique Customers, Total Sales and Total Profit KPI cards. Its visuals show the Top 10 Products by Sales, Top 10 Customers by Sales, Profit by Sub-Category and Sales by Customer Segment.
Year, Market and Category slicers support focused analysis.

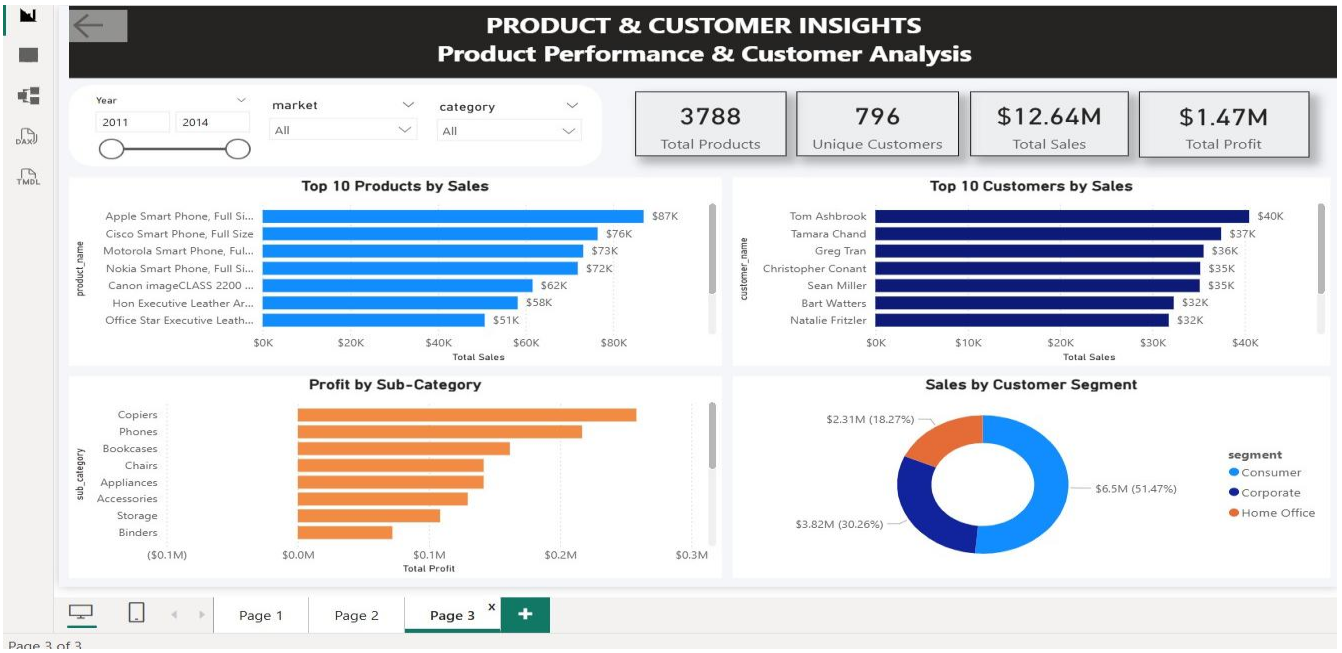


Figure 3. Product & Customer Insights dashboard page.

7. Dashboard Features and Business Insights

Dashboard feature checklist

Required feature	Implementation in the report
Slicers / filters	Year and relevant Market, Region, Segment or Category filters on the three pages.
Interactive visuals	Charts are designed to respond to filter selections and cross-highlighting; interactions should be tested in the PBIX before submission.
KPI cards	Sales, profit, orders, discount, quantity, average order value, products and customers.
Proper formatting	Page titles, visual titles, labels, aligned cards, clear spacing and readable chart layouts.
Consistent colour theme	Navy/blue headings and sales visuals, orange profit/category accents, light neutral page backgrounds.

Key business insights

1. **Sales concentration:** The Central region generated the highest regional sales at $2,821,740. APAC led the market comparison at $3,581,785. These areas are useful starting points for studying what drives stronger sales.

2. **Category profitability:** Technology contributed the largest total category profit at $663,200. Its higher contribution makes it an important category to monitor for sustained profitability.

3. **Customer contribution:** The Consumer segment generated the highest segment sales at $6,503,476. Customer-focused promotions and retention analysis can be prioritized for this segment.

4. **Product performance:** Apple Smart Phone, Full Size was the leading product by sales ($86,936), while Canon imageCLASS 2200 Advanced Copier led the product profit ranking ($25,199.93). Sales leadership and profit leadership are not necessarily the same.

5. **Discount and margin attention:** Furniture had the highest average discount (16.80%) among the three categories. Discount levels should be reviewed alongside profit rather than judged on sales volume alone.

6. **Loss review:** 7,538 distinct orders included at least one negative-profit record. Reviewing the affected products, discount levels, shipping costs and order mix may help identify improvement opportunities.

7. **Shipping pattern:** Standard Class accounted for 30,764 rows, making it the most commonly used shipping mode in the dataset.

8. Conclusion and Submission Readiness

This project demonstrates an end-to-end analytics workflow: preparing a transaction dataset, summarizing performance in Excel, framing and answering business questions with SQL, and communicating the results through a three-page Power BI dashboard. The work provides a structured view of revenue, profit, order activity, product performance, customer contribution, geography, discounts and shipping patterns.

The dashboard aligns with the requested structure in the internship brief and includes slicers, KPI cards, interactive visual design, page navigation and a consistent colour theme. The Excel analysis workbook contains the cleaned data and summary outputs used for the analysis.

Final submission checklist from the internship brief

Required file	Status to confirm before ZIP submission
Cleaned dataset (.xlsx or .csv)	Use the final cleaned dataset file.
SQL queries file (.sql or .pdf)	Include the SQL statements used for the ten required analyses.
Excel analysis file	Include the cleaned dataset and analysis/summary sheets.
Power BI dashboard (.pbix)	Include the saved three-page Power BI report.
Final business insights report (.pdf)	Include this report.

Data note

This report uses the cleaned Excel workbook and the dashboard screenshots available for this project. The numeric summaries were recalculated from the cleaned dataset. The SQL section reports dataset-derived results corresponding to the required query topics; the report does not independently verify the contents of a separately saved SQL script or the live behaviour of every Power BI interaction. Those should be checked in the final project files before submission.

End of Report